

GIT SETUP & PUSH GUIDE

A Beginner's Reference for Version Control with GitHub

Prepared by

Shahid Shaikh

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13

Setup Steps

SSH

Auth Setup

9

Key Commands

Overview

This guide walks you through setting up Git on your local machine, connecting it to a GitHub account, and pushing your first commit to a remote repository. Follow the steps sequentially for a smooth first-time setup.

Part 1 — Installation & Configuration

1 Verify Git Installation

Check whether Git is already installed on your system by running the following command in your terminal:

```
git --version
```

Expected Output:

```
git version 2.x.x
```

NOTE: If Git is not installed, download and install it from <https://git-scm.com> before proceeding.

2 Configure Git Username

Set the display name that will appear in your commit history. Use your full name or preferred alias.

```
git config --global user.name "Shahid Shaikh"

# Verify
git config --global user.name
```

Expected Output:

```
Shahid Shaikh
```

3 Configure Git Email

Use the same email address associated with your GitHub account.

```
git config --global user.email "your-email@example.com"

# Verify
git config --global user.email
```

Expected Output:

```
your-email@example.com
```

4 Verify All Git Configuration

Display all currently configured Git settings to confirm your setup is correct.

```
git config --global --list
```

Expected Output:

```
user.name=Shahid Shaikh
user.email=your-email@example.com
```

Part 2 — Repository Setup

5 Clone GitHub Repository

Clone an existing repository from GitHub to your local machine. Replace <username> and <repository> with your actual values.

```
git clone https://github.com/<username>/<repository>.git

# Example
git clone https://github.com/shahidshaikh/demo-repo.git

# Move into the cloned directory
cd demo-repo
```

6 Verify Remote Repository

Confirm that the local repository is connected to the correct GitHub remote.

```
git remote -v
```

Expected Output:

```
origin  https://github.com/<username>/<repository>.git (fetch)
origin  https://github.com/<username>/<repository>.git (push)
```

7 Check Current Branch

Verify you are on the correct working branch.

```
git branch

# If branch is named 'master', rename it to 'main'
git branch -M main
```

Expected Output:

```
* main
```

Part 3 — Your First Commit & Push

8 Create a Test File

```
echo "Hello Git" > test.txt

# Check status
git status
```

Expected Output:

```
Untracked files:
  test.txt
```

9 Stage Your Changes

Add all modified or new files to the staging area.

```
git add .

# Verify files are staged
git status
```

Expected Output:

```
Changes to be committed:
  new file:   test.txt
```

10 Commit Your Changes

Create a commit with a descriptive message explaining what the change does.

```
git commit -m "Initial commit"

# View commit history
git log --oneline
```

Expected Output:

```
abc1234 Initial commit
```

11 Push Code to GitHub

Push your committed changes to the remote repository on GitHub.

```
git push -u origin main

# For all future pushes, simply use:
git push
```

Expected Output:

```
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

12 Pull Latest Changes

Always pull the latest changes from the remote before starting new work.

```
git pull

# Or explicitly:
git pull origin main
```

Part 4 — Daily Git Workflow

Follow this sequence every time you work on a project to keep your local and remote repositories in sync.

```
# 1. Check the current state of your working directory
git status

# 2. Pull the latest changes from the remote
git pull

# 3. Stage all your changes
git add .

# 4. Commit with a meaningful message
git commit -m "Added new feature"

# 5. Push changes to GitHub
git push
```

Part 5 — SSH Authentication (Recommended)

GitHub no longer supports password-based authentication for Git operations. SSH key authentication is the recommended approach for secure, password-free access.

1 Generate SSH Key

```
ssh-keygen -t ed25519 -C "your-email@example.com"

# Press Enter for all prompts to use default settings
```

2 Start the SSH Agent

```
eval "$(ssh-agent -s)"
```

3 Add Key to SSH Agent

```
ssh-add ~/.ssh/id_ed25519
```

4 Copy Your Public Key

```
# Display your public key – copy the entire output  
cat ~/.ssh/id_ed25519.pub
```

5 Add Key to GitHub

Navigate to GitHub and add the copied key:

- Log in to GitHub
- Go to Settings
- Select SSH and GPG Keys
- Click New SSH Key
- Paste the copied key and click Save

6 Test SSH Connection

```
ssh -T git@github.com
```

Expected Output:

```
Hi <username>! You've successfully authenticated.
```

7 Update Remote URL to Use SSH

```
# Check existing remote  
git remote -v  
  
# Switch from HTTPS to SSH  
git remote set-url origin git@github.com:<username>/<repository>.git  
  
# Confirm the change  
git remote -v
```

Expected Output:

```
origin  git@github.com:<username>/<repository>.git (fetch)
origin  git@github.com:<username>/<repository>.git (push)
```

Part 6 — Troubleshooting

Authentication Failed

If you see an authentication error, try the following solutions:

- Configure SSH authentication following Part 5 of this guide
- Or generate a Personal Access Token (PAT) on GitHub and use it in place of a password

src refspec main does not match any

This error occurs when you try to push without any commits. Ensure you have staged and committed at least once:

```
git add .
git commit -m "Initial commit"
git push -u origin main
```

Updates were rejected

The remote branch has changes not present in your local branch. Rebase before pushing:

```
git pull origin main --rebase
git push origin main
```

Quick Reference — Essential Commands

The most commonly used Git commands for everyday version control:

Command	Description
<code>git status</code>	Show the working tree status
<code>git add .</code>	Stage all changed files

<code>git commit -m "msg"</code>	Commit staged changes with a message
<code>git push</code>	Push commits to the remote repository
<code>git pull</code>	Fetch and merge changes from remote
<code>git branch</code>	List all local branches
<code>git checkout <branch></code>	Switch to the specified branch
<code>git log --oneline</code>	View a condensed commit history
<code>git remote -v</code>	Display remote repository connections

End of Document

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